Machine Learning in Healthcare Sector: Algorithms, Applications, Challenges and Prospects

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Abstract. *Machine learning (ML) algorithms are known for their ability to analyse large amounts of data, uncover exciting relationships, provide interpretation, and identify patterns. In recent years, ML techniques have been widely applied across various fields to address complex challenges, including in healthcare where they have created exciting new opportunities. This study presents a review of the application of ML algorithms (supervised, semi-supervised, unsupervised, and reinforcement learning) in healthcare, with a focus on identifying and summarising the reported ML algorithms being used in diverse domains in the healthcare sector, such as disease detection and diagnosis, drug discovery and development, medical image processing, robotic assisted surgery, DTx, EHR, clinical trials and research, outbreak prediction, etc. The paper also presents the benefits, challenges, and potential of ML-based algorithms in the healthcare industry.*

Keywords: *Machine Learning, Healthcare, Disease Detection and Diagnosis, Drug Discovery and Development, Medical Image Processing, Robotic Assisted Surgery, DTx, EHR, Clinical Trials and Research, Outbreak Prediction.*

1. INTRODUCTION

Machine Learning (ML) refers to the ability of a machine to mimic intelligent human behaviour in order to perform complex tasks. It utilises algorithms that are trained on data sets to develop self-learning models, instead of a rule-based system. These models can predict outcomes and classify information without the need for human intervention. Machine learning (ML) is an interdisciplinary field that encompasses the tools, methods, and techniques used across various domains, including finance, retail, e-commerce, cybersecurity, environment, remote sensing image classification, and speech recognition [Pal and Deswal, 2008; Deswal and Pal, 2008; Kumar and Deswal, 2020; Shehab *et al.*, 2022]. Recently, advancements in ML have created exciting new opportunities in the healthcare sector [Furizal *et al.*, 2023].

Healthcare is the improvement of health via the prevention, diagnosis, treatment, amelioration or cure of disease, illness, injury, and other physical and mental impairments in people. It refers to every aspect, service, process, product, and device which offers care. The main aims of healthcare are improving the quality of care and patient outcomes whilst reducing the cost of care [Mold, 2017]. Healthcare is fundamental to society and is essential for a productive workforce, stable family structure, and basic human happiness [Bitterman, 2007]. It is one of the largest and most important industries in any country, consuming anywhere from 3 to nearly 18 percent of GNP, and still growing [Tulchinsky and Varavikova, 2014]. Global spending on health has increased from US$4.1 trillion (10% of the global economy) in 2004 to US$9.8 trillion (10.3% of global GDP) in 2021 [Edejer *et al.*, 2008; WEC, 2024]. Despite such huge expenses on healthcare, it is not necessarily translating into better health outcomes. The increasing healthcare costs are putting immense strain on households, governments, insurers and healthcare providers, and are becoming unsustainable. However, technology-enabled smart healthcare using ML techniques can play a critical role in different healthcare domains.

ML can be effectively used to analyse important clinical parameters. This includes extracting medical information, predicting diseases and their stages of development, planning and monitoring patients' conditions, precise resource allocation, and enhancing healthcare efficiency [Shehab *et al.*, 2022]. Additionally, ML can assist in data analysis and provide intelligent alerts when necessary [Gharaibeh *et al.*, 2022]. This review paper provides an overview of machine learning (ML) and various ML algorithms based on their data-learning approaches. It highlights the applications of ML in the healthcare sector and offers a comprehensive summary of the techniques employed across diverse domains in healthcare. Additionally, the paper discusses the benefits, challenges, and potential of ML-based algorithms in the healthcare industry.

2. OVERVIEW OF MACHINE LEARNING

The term "*machine learning*" was coined by Arthur Samuel in 1959. It is a discipline of artificial intelligence (AI) that provides machines (computers) the ability to automatically learn from data and past experiences to identify patterns and make predictions with minimal human intervention or explicit programming. In the context of healthcare, ML enables computers to process healthcare (medical) data, identify health patterns and make predictions without explicitly programmed.

The general ML architecture requires several successive processes/stages/steps/components, such as Data collection → Data pre-processing → Choosing the right model → Training the model → Evaluating the model → Hyper-parameter tuning and optimization → Predictions and deployment [Saboor *et al.*, 2022; An *et al.*, 2023]. The data quality and training processes are at the core of ML as the quality of training data significantly impacts the model's performance [Furizal *et al.*, 2023]. The performance (accuracy) of the ML model (algorithm) after being employed in real situations is assessed/evaluated in terms of sensitivity and specificity by using various metrics, such as F1-score, confusion matrix, mean squared error (mse), root mean squared error (rmse), mean absolute error (mae), cross-validation, bias-variance trade-off, etc. [Furizal *et al.*, 2023].

2.1 Types of ML Algorithms

ML algorithms are broadly categorized into four major types based on the type of learning approaches, namely Supervised Learning (SL), Semi-supervised Learning (SSL), Unsupervised Learning (UL), and Reinforcement Learning (RL). The taxonomy of ML algorithms is represented in Fig.1.

2.1.1 Supervised Learning (SL)

It is used in training classification and regression (prediction) algorithms based on previous examples. In this learning technique, the training data set involves known features and corresponding predictions (outcomes), that is referred to as labelled data. In healthcare, labelled data means the data containing patient examples and corresponding labels, such as diagnoses or treatment outcomes. In this approach, the information from the labelled training data set's features is generalised to construct a model that can correctly predict training-set outcomes, and then uses the learned model to make predictions using the new features in the testing data set [Habehh and Gohel, 2021; Furizal *et al.*, 2023]. Some of the widely used SL machine learning classification algorithms are Random Forest (RF), Support Vector Machine (SVM), Neural Networks (NN), Convolutional Neural Networks (CNN), Naïve Bayes, K-Nearest Neighbors (K-NN), and Decision Trees (DT); and regression algorithms are linear regression, logistic regression, multiple linear regression (MLR), Support

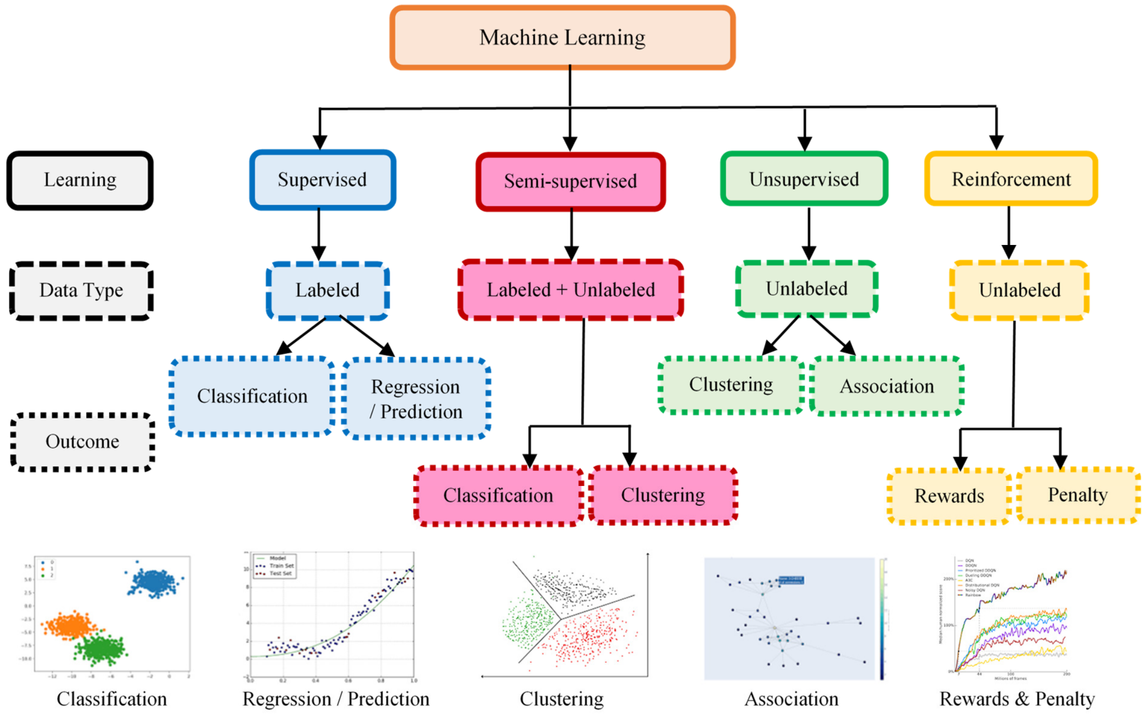

Fig. 1: Taxonomy of machine learning algorithms.

Vector Machine (SVM), Back-Propagation Neural Networks (BPNN), Convolutional Neural Networks (CNN) and Partial Least Squares (PLS). The performance is commonly evaluated by accuracy, sensitivity and specificity in the case of SL classification; and in terms of mean squared error (mse), root mean squared error (rmse), mean absolute error (mae), mean absolute percentage error (mape) in case of regression problems.

2.1.2 Unsupervised Learning (UL)

It is typically used to evaluate large amounts of unlabelled data with highly non-linear learning, and to group data that has not been classified into independent clusters or associations. This method allows for the extraction of features and explores possibilities of data clusters by identifying the underlying relationships or features in the data, and then grouping them by their similarities [Alloghani *et al.*, 2020]. UL is usually purposeful in data analysis, stratification, and reduction rather than prediction. Some of the UL machine learning clustering algorithms used in healthcare are K-means, K-medoids, Hierarchical Clustering (HC), Deep Belief Networks (DBN) and Self Organizing Map (SOM); along with dimensionality reduction algorithms such as Density-based Spatial Clustering of Application with Noise (DBSCAN) and Feature Selection. However, these are semi-popular in healthcare.

2.1.3 Semi-supervised Learning (SSL)

It falls in between SL and UL. This approach uses a small amount of labelled data and a large amount of unlabelled data to train a model. The goal of semi-supervised learning is to learn a function that can accurately predict the output variable based on the input variables, similar to SL. However, unlike SL, the algorithm is trained on a dataset that contains both labelled and unlabelled data. SSL is particularly useful when there is a large amount of unlabelled data available, but it's too cumbersome to label all of it. It is usually employed for the classification and clustering of data. Some of the SSL machine learning algorithms used in healthcare are Transductive Support Vector Machines (TSVM) and Co-Forest.

2.1.4 Reinforcement Learning (SSL)

It is neither supervised nor unsupervised learning, instead based on trial-and-error learning. It is considered as the closest form of learning as seen in humans and animals, which is real-world learning by observing their environment [Sutton and Barto, 2018]. In this approach, algorithms learn through repeated interactions with their environment (that is, a specific problem space). RL algorithms learn to take actions that optimise a goal, such as maximising rewards provided by the environment. It is typically used in decision-making. Some of the RL machine learning algorithms used in healthcare are Recurrent Neural Networks (RNN), Feedforward Neural Networks (FNN), Q-learning, R-learning, Deep Adversarial Networks (DAN), Temporal Difference (TD), and Markov Decision Process (MDP). However, they have limited application in healthcare.

2.1.5 Other Specialized Types of Learning

In addition to the above-discussed major types, there are other variants or specialized types of ML algorithms. These include Transfer Learning (reusing a pre-trained model for a related task), Neural Networks with multiple layers for complex pattern recognition, Ensemble Learning (combining multiple models), NLP algorithms, Deep Learning, etc.

3. APPLICATIONS OF ML IN HEALTHCARE

The number of ML-based applications in healthcare are growing and are seeing gradual acceptance. Some of its major applications in healthcare [TNMLAH, 2023; Chirag, 2024; TTAMLH, 2025] are shown in Fig. 2 and discussed hereunder.

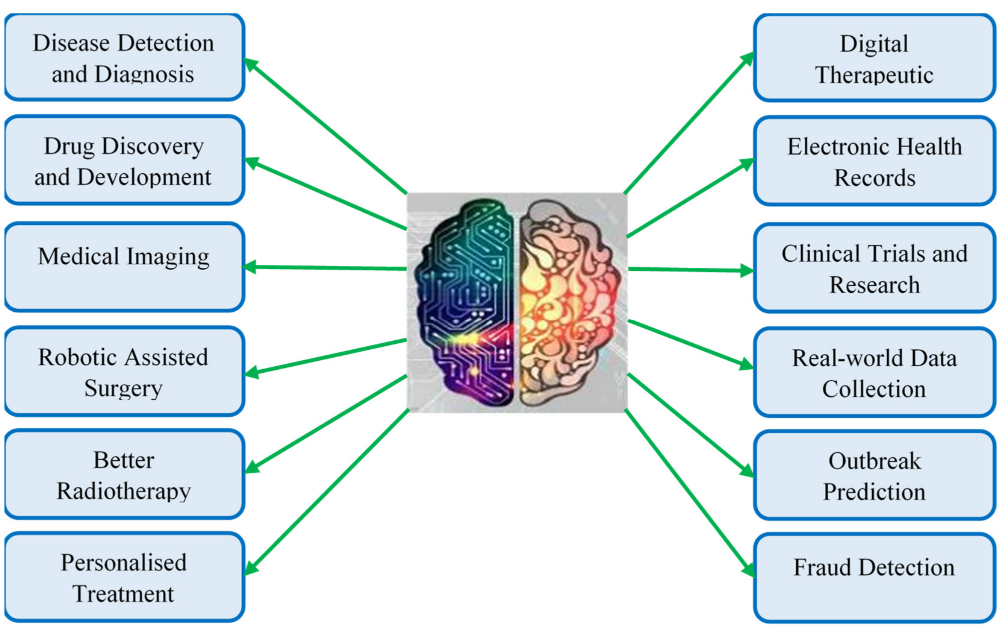

Fig. 2: Application of machine learning (ML) in healthcare.

3.1 Disease Detection and Diagnosis

One of the main applications of machine learning (ML) in healthcare is the early detection and diagnosis of diseases, particularly those that are challenging to identify in their initial stages. This includes identifying conditions such as malignancies that are often difficult to catch early, as well as genetic disorders, heart attacks, sepsis, and other serious health issues. Early disease detection is crucial for timely and effective treatment, helping patients maintain a good quality of life.

These technologies utilise information from previous patient records and daily assessments. Additionally, they incorporate real-time measurements of vital signs such as heart rate and blood pressure. ML technologies can alert healthcare personnel to potential risks for patients, allowing them to take preventive measures. Furthermore, machine learning aids in treating clinical diseases by automating as much of the process as possible.

3.2 Drug Discovery and Development

ML in healthcare can also help in drug discovery and development by making the process faster and more cost-effective. This includes research and development technologies such as next-generation sequencing and precision medicine, which can help identify alternative treatment pathways for multifactorial diseases. Currently, ML techniques utilise unsupervised learning to identify patterns in data without making specific predictions.

3.3 Medical Imaging

Medical imaging analytics represents one of the most remarkable applications of ML in the medical field. It is extensively utilised in hospitals and clinics to identify abnormalities and detect subtle changes in medical images – such as MRI, X-ray, CT scans, PET scans, ultrasound, and other radiology scans. This technology helps in the early detection and diagnosis of potential health risks through the use of a radiology information system. Image recognition plays a crucial role in assisting doctors in identifying tumours, improving cancer prognosis, diagnosing liver and kidney infections, and much more.

3.4 Robotic Assisted Surgery

As one of the most successful applications of ML in healthcare, ML-powered surgical robots have transformed surgical procedures by enhancing both accuracy and speed, even in the most challenging situations. These robotic systems utilise ML algorithms to improve the precision of surgical tools by incorporating real-time measurements during surgery. As a result, they can perform complex surgical procedures with a reduced risk of blood loss, side effects, and pain. Furthermore, patients experience a much faster and easier recovery after surgery.

3.5 Better Radiotherapy

One of the most sought-after applications of ML-based techniques in healthcare is in the area of radiology. Medical image analysis involves many discrete variables that can arise at any particular moment in time. There are many lesions, cancer foci, etc. which cannot be simply modelled using complex equations. However, ML-based algorithms can easily categorise them as normal or abnormal, lesion or non-lesion, benign or malignant, etc. It is due to the inherent capability of these algorithms to learn from the multitude of different samples available on-hand that enables them to diagnose and find the variables easily.

3.6 Personalised Treatment

Medical treatments are most successful when they take individual needs into account. One of the primary applications of machine learning (ML) in healthcare is the provision of personalised treatments. These tailored approaches can be more effective by combining individual health data with predictive analytics. Additionally, this area is open to further research and can lead to improved disease assessment.

Currently, physicians face limitations in their ability to select from a narrow range of diagnoses and must estimate a patient’s risk based on their symptomatic history and available genetic information. However, ML offers a way for physicians to provide personalised care by analysing a patient's medical history, symptoms, and test results. This technology can generate multiple treatment options and develop customised treatments, allowing for the prescription of medications that target specific diseases in each patient for faster recovery. Additionally, ML can assist doctors in determining whether a patient is ready for necessary changes in their medication.

3.7 Digital Therapeutic (DTx)

Behavioural modification is a crucial aspect of preventive medicine. With the rise of machine learning in healthcare, numerous startups are emerging in areas like cancer prevention, identification, and patient treatment. ML-based algorithms can analyse various types of patient data, including medical history, genetic information, and lifestyle factors. By continuously monitoring and adjusting treatment plans, ML-powered digital therapeutics enhance treatment effectiveness while minimising adverse effects. Further, there are ML-based apps that recognize gestures that people make in their daily lives, which helps in understanding unconscious behaviour and suggests necessary changes.

3.8 Electronic Health Records (EHR)

Maintaining up-to-date health records is often a time-consuming and expensive task. ML-based techniques have the potential role in the healthcare sector by streamlining processes, thereby saving time, effort, and money. Document classification methods using ML-based techniques represent the next generation of intelligent electronic health records (EHRs). The smart health record systems integrate ML tools from the ground up to assist with diagnosis, clinical treatment recommendations and other healthcare functions.

Prescription errors are among the most serious medical mistakes, often leading to patient deaths. These errors can occur when a healthcare provider prescribes the wrong medication, incorrect dosage, or fails to consider potential drug interactions. To help reduce these types of medical mistakes, ML solutions in healthcare can be a valuable resource. Machine learning models in healthcare analyse historical EHR data and compare new prescriptions against these records. Prescriptions that deviate from typical patterns are flagged for doctors to review and adjust. Brigham and Women’s Hospital employs an ML-based system to identify prescription errors. Over the course of a year, this system detected 10,668 errors, with 79% of them deemed clinically significant [Chirag, 2024]. As a result, the hospital was able to save $1.3 million in healthcare costs. In addition to cost savings, the ML-powered error detection system enhances the quality of care by preventing drug overdoses and reducing health risks.

3.9 Clinical Trials and Research

Clinical trials and research play a crucial role in ensuring the safety and effectiveness of medical solutions. However, these processes can be costly and lengthy, often taking years to complete. In urgent situations, such as the need for COVID-19 vaccines, it is essential to expedite the release of these solutions. ML algorithms in healthcare can help shorten this timeline. ML-based predictive analytics help in identifying potential clinical trial candidates from a broad range of data sources, including past doctor visits and social media activity. Additionally, ML techniques allow for real-time monitoring and data access for trial participants, facilitate the analysis of ongoing data, and help reduce inaccuracies by leveraging EHRs.

3.10 Real-world Data (RWD) Collection

Real-world data (RWD) in the medical and healthcare field "are the data relating to patient health status and/or the delivery of health care routinely collected from a variety of sources" [USFDA, 2025]. Examples of RWD include data derived from electronic health records, medical claims data, data from product or disease registries, and data gathered from other sources (such as digital health technologies) that can inform on health status. Crowdsourcing has become increasingly popular in the medical field, allowing researchers and practitioners to access a vast amount of information contributed by individuals with their consent. This live health data has significant implications for the future of medicine. With advancements in the Internet of Things (IoT) and machine learning (ML) techniques, the healthcare sector continues to explore new ways to utilize this data. These innovations can help tackle difficult-to-diagnose cases and improve overall diagnosis and medication.

3.11 Outbreak Prediction

ML techniques are used by healthcare organisations to monitor and predict epidemics and pandemics worldwide. Today, scientists have access to vast amounts of data gathered from various sources, including satellites, real-time social media updates, and websites. ML techniques help analyse this information and predict outbreaks of diseases ranging from malaria to severe chronic infections. These predictions are especially beneficial for developing countries, as they often lack essential medical infrastructure and educational resources.

3.12 Fraud Detection

The estimated losses from fraud and corruption in healthcare were about $455 billion of the $7.35 trillion spent globally on healthcare in 2018, according to the National Academies of Sciences, Engineering, and Medicine [Timofeyev and Jakovljevic, 2022]. By utilizing ML in healthcare, businesses can identify invalid claims and suspicious behaviour before they are processed for payment. This technology also accelerates the approval and payment of valid claims. In addition to detecting insurance fraud, machine learning can help prevent the theft of patient data.

4. ML ALGORITHMS APPLIED IN DIVERSE DOMAINS OF HEALTHCARE SECTOR

Given the several types of ML techniques, it is highly imperative to identify and implement an approach/technique suitable and specific to the particular healthcare application. Several factors, including the number of features, sample size and data distributions, can have significant effects on the learning and prediction processes and should be considered. Based on an in-depth review of over 125 research publications, the utility and potential of specific ML algorithms across diverse healthcare domains have been tabulated in Table 1 for ready reference.

Table 1: Summary of ML techniques applied in diverse domains of healthcare.

S. No.	Healthcare Domains	ML Technique	Reference
1	Disease Detection and Diagnosis	K-NN	Shouman *et al.* (2012), Deekshatulu and Chandra (2013), Shimpi *et al.* (2017), Enriko *et al.* (2018).
		EC	Huda *et al.* (2016).
		DT	Maji and Arora (2019), Pathak and Valan (2020).
		NB	Cheung (2001), Gupta *et al.* (2019), Repaka *et al.* (2019), Gupta *et al.* (2020), Balaha and Hassan (2023).
		SVM	Shimpi *et al.* (2017), Lopez-Ubeda *et al.* (2021).
		TSVM	Filipovych and Davatzikos (2011).
		RF	Shimpi *et al.* (2017), Lopez-Ubeda *et al.* (2021).
		CNN	Urban *et al.* (2018), Ali *et al.* (2020), Rasheed *et al.* (2020), Fernando *et al.* (2021), Lopez-Ubeda *et al.* (2021), Goel *et al.* (2022).
		RNN	Rasheed *et al.* (2020), Fernando *et al.* (2021).
		LR	Shimpi *et al.* (2017).
		SOM	Zhang *et al.* (2010), Mishra and Panda (2018).
2	Drug Discovery and Development	DT	Maji and Arora (2019), Wu and Hicks (2021).
		NB	Repaka *et al.* (2019).
		CNN	Lin and Wong (2018).
		SVM	Heikamp and Bajorath (2014), Wu and Hicks (2021).
		K-NN	Wu and Hicks (2021), Obaido *et al.* (2024).
		SOM	Xiao *et al.* (2005), Schneider *et al.* (2009).
3	Medical Imaging	SVM	Boero *et al.* (2017), Ali *et al.* (2019), Lopez-Ubeda *et al.* (2021), Michael *et al.* (2022).
		RF	Lopez-Ubeda *et al.* (2021), Michael *et al.* (2022).
		K-NN	Michael *et al.* (2022), Gupta *et al.* (2023).
		CNN	Gupta *et al.* (2023), Akbari *et al.* (2018), Dhillon *et al.* (2019), Retson *et al.* (2019), Esteva *et al.* (2019), Ker *et al.* (2019), Wang *et al.* (2020b), Fernando *et al.* (2021), Lopez-Ubeda *et al.* (2021), Sarvamangala and Kulkarni (2021), Velden *et al.* (2022), Muis *et al.* (2023).
		RNN	Esteva *et al.* (2019), Castiglioni *et al.* (2021), *Fernando et al.* (2021), Agarwal *et al.* (2024).
		DBN	Faturahman *et al.* (2017), Kaur and Singh (2020).
		DT	Patel *et al.* (2015), Zhou *et al.* (2015), Drukker *et al.* (2024).
		TSVM	Filipovych and Davatzikos (2011), Zemmal *et al.* (2016).
		SOM	Chang and Teng. (2007), Mishra and Panda (2018).
4	Robotic Assisted Surgery	RNN	DiPietro *et al.* (2019), Xiaoguo *et al.* (2019), Wu *et al.* (2021).
		FNN	Thuruthel *et al.* (2016), Grassmann *et al.* (2018).
		CNN	Xiaoguo *et al.* (2019), Ahmad *et al.* (2020).
		Q-learning	Sumanas *et al.* (2022), Gao (2024).
		DAN	Islam *et al.* (2019), Nema and Vachhani (2023).
		MDP	Fakoor *et al.* (2016), Guo *et al.* (2024).

Table 1: *Contd.*

S. No.	Healthcare Domains	ML Technique	Reference
5	Better Radiotherapy	CNN RNN	Saba *et al.* (2019), Khan *et al.* (2020), Nakaura *et al.* (2020), Qian and Fang (2022). Saba *et al.* (2019), Nakaura *et al.* (2020).
6	Personalised Treatment	NB RT CNN SVM RF LR NN	Karlik (2012), Chakrapani *et al.* (2021), Ma *et al.* (2020). Tang *et al.* (2017), Zhan *et al.* (2021). Chieregato *et al.* (2022), Kirelli *et al.* (2023). Zhan *et al.* (2021), Yu *et al.* (2022). Zhan *et al.* (2021), Zhu *et al.* (2022), Yu *et al.* (2022). Yu *et al.* (2022). Fu *et al.* (2012), Yu *et al.* (2022), Parekh *et al.* (2023).
7	Digital Therapeutic	NN K-NN SVM DT CNN RNN	Bucher *et al.* (2024). Hu *et al.* (2024). Bucher *et al.* (2024). Chen *et al.* (2019), Carrera *et al.* (2024). Yoo *et al.* (2023), Wang *et al.* (2025). Nigo *et al.* (2022), Yoo *et al.* (2023).
8	Electronic Health Records (EHR)	RNN SVM DT CNN LR RF	Bekhet *et al.* (2018), Ge *et al.* (2019), Rasmy *et al.* (2021). Wu *et al.* (2010), Liang *et al.* (2014), Hurst *et al.* (2022). Liang *et al.* (2014), Hurst *et al.* (2022). Liu *et al.* (2018), Chieregato *et al.* (2022). Rahman *et al.* (2022), Saraswat *et al.* (2024). Hurst *et al.* (2022), Saraswat *et al.* (2024).
9	Clinical Trials and Research	RNN SVM NN RF SOM LoR NB K-NN	Moreau *et al.* (2018). Wang *et al.* (2020b), Battanova *et al.* (2023). Wang *et al.* (2020b). Kalscheur *et al.* (2018), Ozkanca *et al.* (2019), Bedon *et al.* (2022), Wang *et al.* (2020a), Battanova *et al.* (2023). Parra-Rodríguez *et al.* (2022). Wang *et al.* (2020b), Battanova *et al.* (2023). Wang *et al.* (2020b). Mansor *et al.* (2012), Wang *et al.* (2020b).
10	Real-world Data Collection	RNN SVM DT CNN RF	Bekhet *et al.* (2018), Rasmy *et al.* (2021). Wu *et al.* (2010), Liang *et al.* (2014), Nordin *et al.* (2020), Hurst *et al.* (2022), Sharin *et al.* (2022). Liang *et al.* (2014), Maji and Arora (2019), Pathak and Valan (2020), Hurst *et al.* (2022). Liu *et al.* (2018), Urban *et al.* (2018), Chieregato et al. (2022). Wang *et al.* (2020a), Galassso *et al.* (2022), Hurst *et al.* (2022).
11	Outbreak Prediction	SVM RF	Nordin *et al.* (2020), Sharin *et al.* (2022), Cho *et al.* (2023). Wang *et al.* (2020a), Galassso *et al.* (2022), Cho *et al.* (2023).
12	Fraud Detection	DT RF LoR SVM ANN SOM	Gharehchopogh *et al.* (2012), Sailaja *et al.* (2021). Nabrawi and Alanazi (2023), Lakshmi *et al.* (2024). Sailaja *et al.* (2021), Nabrawi and Alanazi (2023). Francis *et al.* (2011), Kirlido and Asuk (2012). Lamyo *et al.* (2018), Nabrawi and Alanazi (2023). Olszewski (2014).

ANN: Artificial Neural Networks; BPNN: Back-Propagation Neural Networks; CNN: Convolutional Neural Networks; DAN: Deep Adversarial Networks; DBN: Deep Belief Networks; DT: Decision Tree; EC: Ensemble Classification; FNN: Feedforward Neural Networks; K-NN: K-Nearest Neighbour; LR: Linear Regression; LoR: Logistic Regression; MDP: Markov Decision Process; MLR: Multiple Linear Regression; NB: Naïve Bayes; NN: Neural networks; PLS: Partial Least Squares; RF: Random Forest; RNN: SOM: Self Organizing Map; SVM: Support Vector Machines; TVSM: Transductive SVM; Recurrent Neural Networks.

5. DISCUSSION

The significance of ML applications in the healthcare sector is profound and wide-reaching, as highlighted in the previous section. The adoption of ML-based applications is increasing at a rapid pace in the healthcare sector as it offers numerous benefits, including assisting doctors in early disease detection, researching new medications, creating personalised treatment plans, saving lives, and reducing costs [TNMLAH, 2023; Chirag, 2024; TTAMLH, 2025]. Some of the key benefits (Fig. 3) are –

- **Accelerated data collection** through IoT-connected medical devices enables real-time data gathering, allowing machine learning to process and adapt quickly.

- Accelerated Data Collection
- Cost Efficient Process
- Predictive Analytics
- Customised Real-Time Treatment Plans/Adjustments
- Enhanced Diagnostic Accuracy
- Fast Drug Discovery and Development

- Data Quality
- Effective Management of Complex Datasets
- Data Privacy and Security
- Risk of Misinterpretation
- Resource Constraints
- Medical Ethics

Fig. 3: Benefits and challenges of ML in healthcare.

- The adoption of ML is a **cost-effective process**, leading to significant cost savings in different healthcare activities/areas. For instance, it can quickly analyse electronic health records (EHRs), manage patient records, schedule appointments, and automate processes, thus reducing time and resources spent on repetitive tasks.
- **Predictive Analytics by ML techniques** is helpful in anticipating patient health deterioration, disease progression, and readmission risks; thereby, allowing proactive interventions to reduce hospital readmissions.
- ML analyses individual patient data to **customise real-time treatment plans and/or adjustments** according to each patient's specific needs and evolving conditions; thus, improving outcomes and reducing adverse effects.
- ML algorithms have **enhanced diagnostic accuracy** due to remarkable precision in scrutinising medical images and patient data that facilitate earlier and more accurate diagnoses of conditions, particularly rare and complex diseases that human observers may overlook.
- Machine learning accelerates the **drug discovery and development** process by analysing extensive datasets to pinpoint potential drug candidates, predict their efficacy, and uncover potential side effects, thus streamlining research and development by reducing associated time and costs.

Although ML-based applications in healthcare provide substantive benefits along with unique and progressive opportunities; however, their adoption also faces significant challenges that need to be addressed (Fig. 3). These include ensuring data quality, effectively managing complex datasets, safeguarding data privacy and security, minimising the risk of misinterpretation, overcoming resource constraints, and navigating medical ethics [Furizal *et al.*, 2023]. Each of these factors critically impacts the effectiveness of machine learning algorithms and cannot be overlooked.

Machine learning (ML) has started exhibiting its potential in the healthcare sector, particularly in improving medication research, patient care, and administrative processes. In the future, ML techniques are expected to complement clinicians rather than replace them, leading to better healthcare quality and a more efficient, cost-effective healthcare system. This progress will benefit patients, providers, insurers, regulators and legislators.

6. CONCLUSIONS

This paper provides an in-depth examination of the application of machine learning (ML) techniques in diverse domains of the healthcare sector, discussing recent studies in the field. Over 125 research publications were reviewed and analysed to showcase the utility and potential of specific ML techniques across diverse healthcare domains. The paper addresses both the benefits and limitations of implementing ML solutions to tackle diverse healthcare challenges. Overall, this review highlights key researchers in the medical field who are utilising ML techniques and aims to serve as a valuable resource for researchers, industry professionals, and policymakers in their future research and application projects.

Declaration of Competing Interest

The author has no known competing financial interests or personal relationships that could have appeared to influence the review work reported in this paper.

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